CLIMATE CHANGE

- **2025: 2nd Warmest Year in 65 years with 23.9 °C average temperature & 3% below average rainfall**
- **IMF RSF Progress with Completion of Four Key Reforms**
- **Pakistan Climate Prosperity Plan (CCP): Climate Investment Blueprint Launched**
- **Pakistan Green Taxonomy (PGT) Issued and Implemented**
- **Pakistan Monsoon 2026 Strategic Plan Prepared**
- **Climate and Disaster Resilience Enhancement Programme (CDREP) Subprogram-II under Implementation: Allocation US$ 500 million**
- **Green Climate Fund Regional Programme G2F Approved: Allocation US$ 27.5 million**

Chapter 17

Climate Change

Climate change has become a defining global challenge, with far-reaching implications for human well-being, economic stability and sustainable development. Widespread and rapid changes have already been experienced in the atmosphere, oceans, cryosphere and biosphere. According to the Intergovernmental Panel on Climate Change Sixth Assessment Report (IPCC AR6)[1], human-induced climate change is intensifying weather and climate extremes in every region of the world. This has caused widespread losses and damages to both natural and human systems. As per the report, approximately 3.3 to 3.6 billion people currently live in highly climate-vulnerable environments, where structural development constraints further amplify exposure and reduce adaptive capacity. Although vulnerable populations contribute the least to global emissions, they are affected the most by climate change. Between 2010 and 2020, mortality caused by floods, droughts and storms is estimated to be fifteen times higher in highly vulnerable regions compared to those with less vulnerability.

The continued greenhouse emissions are expected to drive the global warming to 1.5 °C[2] in the near term (2021-2040). Every additional increase in temperature intensifies concurrent hazards, including compound heatwaves, droughts and extreme precipitation events. These dynamics are driving a more volatile global water cycle, marked by heightened variability and increasingly frequent extremes of both wet and dry conditions. Climate change is further transforming rare and recurring hazards into a 'new normal' of persistent risks, as highlighted in the Climate Risk Index (CRI)[3] 2026. Similarly, Global Risks Report[4] 2026 by World Economic Forum ranked extreme weather events among the top five global risks in the short term (1-2 years) and projected extreme weather events as the highest-ranked long-term risk over a 10-year horizon. Together, these projections establish climate change as a systemic global threat, which requires collective action. Consequently, the human and economic costs of insufficient climate action continue to escalate. According to CRI[5] 2026, more than 832,000 lives were lost and nearly US $ 4.5 trillion in direct economic damages was recorded from over 9,700 extreme weather events between 1995-2024, globally.

Pakistan stands as one of the most severely affected countries. Climate change is no longer a distant or abstract threat to the country but a present reality and an escalating challenge to the country's economy and population. Although Pakistan is a minor contributor to global emissions at less than 1 percent and historic emissions at 0.4 percent[6], the country bears a disproportionately high burden of global climate change. Coupled with this, the country's vulnerability is also linked to its geography, spans from the world's second-highest peaks in

[1]Intergovernmental Panel on Climate Change Sixth Assessment Report, IPCC (IPCC_AR6_SYR_FullVolume.pdf)
[2]Under very high GHG emissions scenario (SSP5-8.5) & intermediate and high GHG emission scenario (SSP2-4.5, SSP3-7.0), Intergovernmental Panel on Climate Change Sixth Assessment Report, IPCC (IPCC_AR6_SYR_FullVolume.pdf)
[3]Climate Risk Index (CRI) 2026, Germanwatch (CRI 26 full report.pdf)
[4]Global Risk Report 2026, World Economic Forum (WEF_Global_Risks_Report_2026.pdf)
[5]Climate Risk Index (CRI) 2026, Germanwatch (CRI 26 full report.pdf)
[6]World Bank (2026), World Development Indicators. Total greenhouse gas emissions excluding LULUCF (MtCO2e) (World Development Indicators | DataBank)

the north to low-lying coastal areas in the south. The Hindu Kush, Himalaya and Karakoram ranges contain the largest non-polar glacial reserves, while the Arabian Sea coastline further amplifies the country's sensitivity to climatic shifts, resulting in wide-ranging and spatially uneven effects of climate change. In the last 50 years, the annual mean temperature in Pakistan has increased by approximately 0.5 °C[7], with projections indicating a further rise by 3 °C to 5 °C by the end of this century. In particular, for the year 2025, northern regions have warmed more intensively, temperature anomalies reached +1.24 °C in Gilgit-Baltistan (GB), +1.29 °C in Khyber Pakhtunkhwa and +1.56 °C in Azad Jammu & Kashmir (AJK), all ranked 1st highest during the past 65 years[8]. These rising temperature records are therefore accelerating glacial melt, increasing the risk of glacier lake outburst floods while intensifying atmospheric moisture, resulting in intensified rainfall variability in the country.

This rainfall variability has been increasingly pronounced due to changing monsoon dynamics of Pakistan. Notably, there has been a decrease in the number of rainy days, accompanied by an increase in rainfall intensity, resulting in more frequent and extreme precipitation events. Additionally, the monsoon patterns are also shifting geographically, with increasing concentration in southern regions, thereby altering flood risk profile of the country. Over the past decade, excessive heat and rainfall have led to several major and minor floods. The 2022 floods appeared as one of the most devastating climate disasters in the country's history. Similarly, the recent 2025 floods emerged as one of the most widespread climate-related disasters, as it hit all four provinces; Punjab, Sindh, Khyber Pakhtunkhwa and Balochistan as well as AJK.

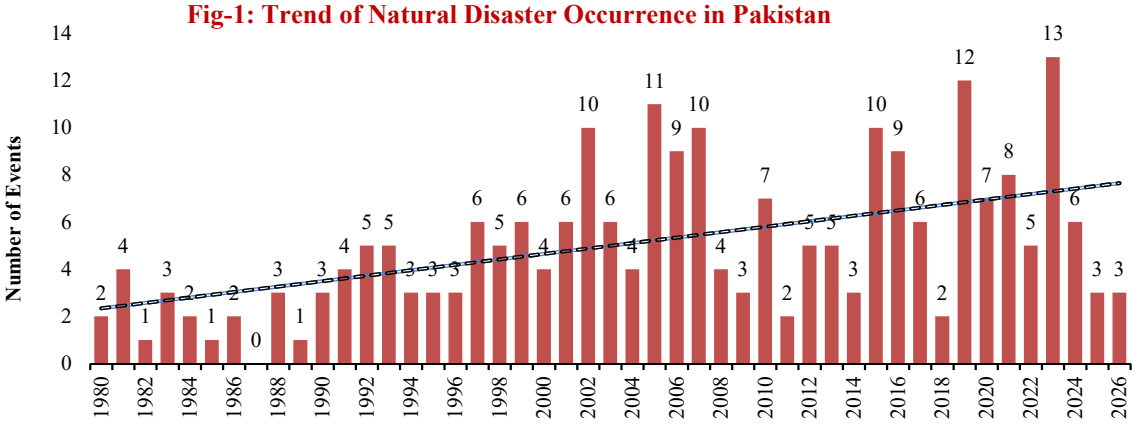
Beyond immediate disaster impacts, climate change has become a systemic risk to Pakistan's economic and social stability. Rising sea levels and saline intrusion in the Indus Delta are degrading coastal ecosystems, including mangroves and fisheries, while increasing cyclonic activity threatens coastal populations. At the same time, deteriorating air quality, climate-affected communities and water scarcity are compounding existing socioeconomic pressures. Thus, Pakistan's climate crisis is not a future threat but a present and escalating reality already unfolding across multiple sectors. With adaptive capacity still developing and high exposure to climate risks, the country remains on the frontline of a global crisis it has contributed little to create.

17.1 Historical Trend of Natural Disaster Occurrence in Pakistan

The trend of natural disaster occurrence in Pakistan during 1980-2026 reflects a clear long-term rise in the frequency of disaster events, highlighting the country's increasing exposure and vulnerability to natural hazards. During the 1980s, the number of recorded disasters remained relatively low, generally fluctuating between 0 and 4 events annually (Figure 1). However, from the early 1990s onward, disaster occurrence began to increase gradually, with more frequent fluctuations in recurring years. A more pronounced escalation in the occurrence of the disaster is visible after 2000, when annual disaster events repeatedly crossed the range of 6 to 11 incidents. Several peak years were recorded during the mid-2000s and, most notably, after 2015, indicating a growing pattern of recurrent natural disasters, including floods, storms, heatwaves, droughts, landslides and other extreme climatic events. The year 2023 proved to the most disaster-prone year in 43 years with 13 natural events recorded, demonstrating the intensified incidence of the disaster risks in the country. The upward-sloping trend line further depicts that natural disasters are becoming more frequent over time rather than isolated or cyclical events thereby highlighting the urgency of strengthened climate resilience, disaster preparedness and integrated risk management.

[7]Climate Change Profile of Pakistan 2017, Asian Development Bank (climate-change-profile-pakistan.pdf)
[8]State of Pakistan Climate Report 2025, Pakistan Metrological Department (Pakistan_Climate_2025.pdf)

Fig-1: Trend of Natural Disaster Occurrence in Pakistan



Source: EM-DAT (The Internatonal Disaster Database)

17.2 Global Climate & Air Quality Snapshot and Pakistan's 2025 Profile

The following presents a snapshot of key climate indicators in 2025 against their typical range of variability under a business-as-usual scenario, along with the assessment of air pollution condition at global level and in Pakistan.

17.2-a State of Global Climate and Air Quality in 2025

The State of the Global Climate[9] 2025 report by World Meteorological Organization (WMO) confirms that 2015-2025 are the hottest eleven years on record, with 2025 ranking as the second or third warmest year in the 176-year observational record, at about 1.43 °C above the 1850-1900 average. This sustained warming is primarily driven by increasing concentrations of heat-trapping greenhouse gases. Among key indicators, the greenhouse gases data show that in 2024 atmospheric carbon dioxide concentrations reached their highest level since its measurements began in 1957 due to continued fossil fuel emissions and reduced effectiveness of land and ocean carbon sinks. This increase along with rise in methane and nitrous oxide gases have upset the Earth's energy balance, particularly over the past two decades, reaching a new high in 2025.

Consistent with this, the indicator of ocean heat content reached a new record high in 2025, exceeding the previous record set in 2024 and of the past nine years. Although 2025 is marked by the shift towards La Niña conditions that temporarily cool the planet, around 90 percent of the ocean surface experienced at least one marine heatwave in 2025. Further, the indicator of global mean sea level, in 2025 remained near the record highs observed in 2024 thereby continued to damage coastal ecosystems and contribute to groundwater salinization and flooding. At the same time, glacier mass loss and declining sea-ice continue to signal rapid cryosphere changes. In the 2024-2025 hydrological year, glacier mass loss was among the five worst on record since 1950. These ocean warming and sea-level rise are projected to continue in the coming centuries and much of this is long-term and irreversible in nature. Further, the world continued to experience extreme events. Thus, the rapid and large-scale changes in the earth system are creating cascading impacts on both human and natural systems thereby, causing food insecurity and displacement, especially in countries with high vulnerability and low adaptive capacity.

While air pollution has long been recognized as a persistent global hazard, but 2025 marked a turning point as the crisis moved to the forefront on the international agenda after being placed at a top ranked global risk by the World Economic Forum's Global Risks Report 2025[10]. The 2025 IQAir World Air Quality Report[11] highlights continued deterioration of global air quality,

[9]State of the Global Climate 2025, World Meteorological Organization (https://wmo.int/publication-series/state-of-global-climate/state-of-global-climate-2025)
[10]Global Risk Report 2025, World Economic Forum (WEF_Global_Risks_Report_2025.pdf)
[11]World Air Quality Report, 2025, IQAir (2025 World Air Quality Report | IQAir)

with wildfire activity severely impacting regions that had historically experienced relatively low PM2.5 levels. Consequently, only 14 percent of cities met the World Health Organization (WHO) annual average PM2.5 guideline of 5 µg/m³ threshold, down from 17 percent in 2024. Similar to a pattern seen in 2024 and in the preceding years, the largest population-weighted PM2.5 concentrations were still concentrated in Africa & Central and South Asia. Central and South Asia remained the top most polluted region with respect to annual average PM2.5 levels as 17 of the 20 most polluted cities located within the region. Taken together, the worsening global air quality alongside intensifying climate pressures underscores brewing environmental crises at global level.

17.2-b State of Pakistan Climate and Air Quality in 2025

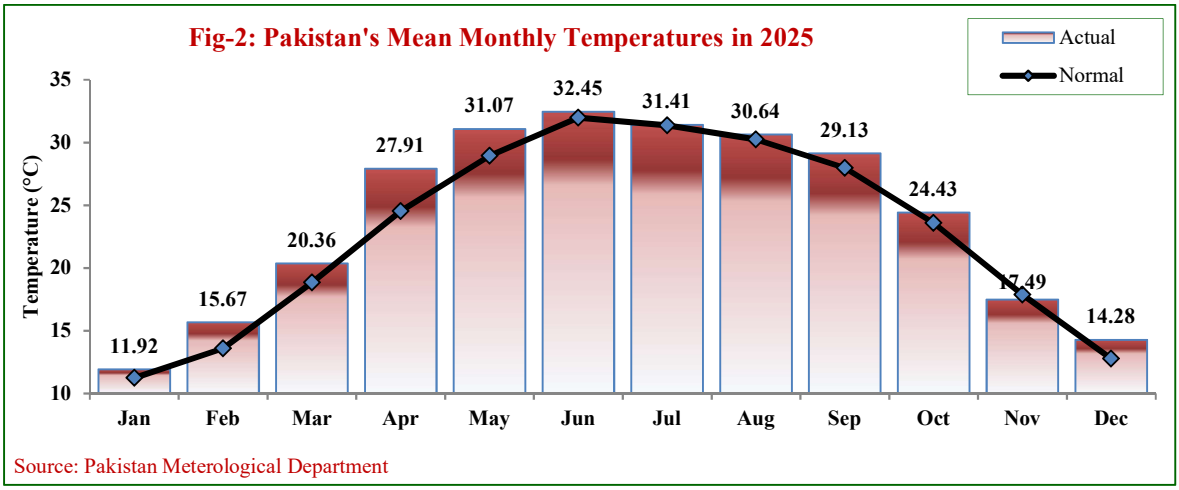
The year 2025 recorded a national annual mean temperature of 23.9 °C in Pakistan, which was 1.09 °C above the long-term average and ranked as the 2nd warmest year on record during past sixty-five years. In contrast, rainfall during the year was near average with a slight decline of 3 percent below the average. Additionally, significant climate events in 2025 reflect notable variations in temperature, rainfall patterns and incidence of wind extremes across Pakistan. According to State of Pakistan Climate Report[12] 2025, during the year, temperature extremes were particularly pronounced in southern regions, where Sibbi (Balochistan) been hottest place recorded the highest average period maximum temperature of 37.8 °C. In contrast, colder conditions prevailed in the northern areas, where Skardu (GB) been coldest place recorded the lowest average period minimum temperature of 7.0 °C. Rainfall distribution also exhibited significant regional disparities. The highest rainfall was recorded in Malam Jabba (Khyber Pakhtunkhwa) at 1461.0 mm, whereas Sialkot (Punjab) experienced the wettest day with 363.3 mm on 27th August 2025. In contrast, Panjgur (Balochistan) remained the driest location, receiving only 0.5 mm of rainfall. Overall, these patterns highlight pronounced spatial contrasts in both temperature extremes and precipitation, reflecting the diverse climatic conditions experienced across the country during 2025. In regards to wind extremes, strong wind events (≥30 knots) were reported extensively across multiple stations nationwide. In particular, very high wind speeds, including peaks of 75 knots experienced at Lahore Airport, 70 knots at Sialkot Airport and Padidan, and 65 knots at Multan Airport, reflecting the widespread occurrence and extreme intensity of winds experienced during the year.

17.2-b (i) Temperature: Another Warm Year

Pakistan experienced national annual 2025 mean temperature of 23.9 °C, was +1.09 °C warmer than average of 22.8 °C (baseline period 1961–1990), and ranked as the 2nd warmest year during past sixty-five years. A similar situation of well above average temperatures prevailed across all the sub-regions: for AJK with 17.77 °C (1.56 °C higher), GB with 15.46 °C (1.24 °C higher) and Khyber Pakhtunkhwa 20.89 °C (1.29 °C more) all ranked 1st highest during the past 65-years (the records are 17.43 °C in 2001, 15.3 °C in 1971 & 20.63 °C in 2022 respectively), Sindh with 27.67 °C (1.09 °C higher) ranked 4th highest (the record is 27.89 °C in 2018), Balochistan with 24.24 °C (1.13 °C more) ranked 5th highest (record 24.99 °C in 2002) and Punjab with 25.35 °C (0.8 °C higher) ranked 9th highest (the record is 25.62 °C in 1970) annual mean temperature.

On a monthly basis, temperature fluctuations are also observed reflecting warming trend in 2025 (Figure 2). January with national mean monthly temperature of 11.92 °C was 0.69 °C warmer than average at country-level. In the month of February national mean temperatures at country-level and across the sub-regions were significantly warmer than average by 2.07 °C higher. Further, the mean temperatures in the pre-monsoon season AMJ 2025 also remained warmer than average by 1.94 °C more than the country average 28.53 °C. Followed by it the summer/ monsoon (JAS 2025) seasonal mean temperatures were warmer than average with a positive anomaly (0.5 °C). Lastly, the year also ended with the warmer temperature at country-level as post-monsoon (OND) seasonal mean temperature appeared warmer than average with a positive anomaly of 0.64 °C.

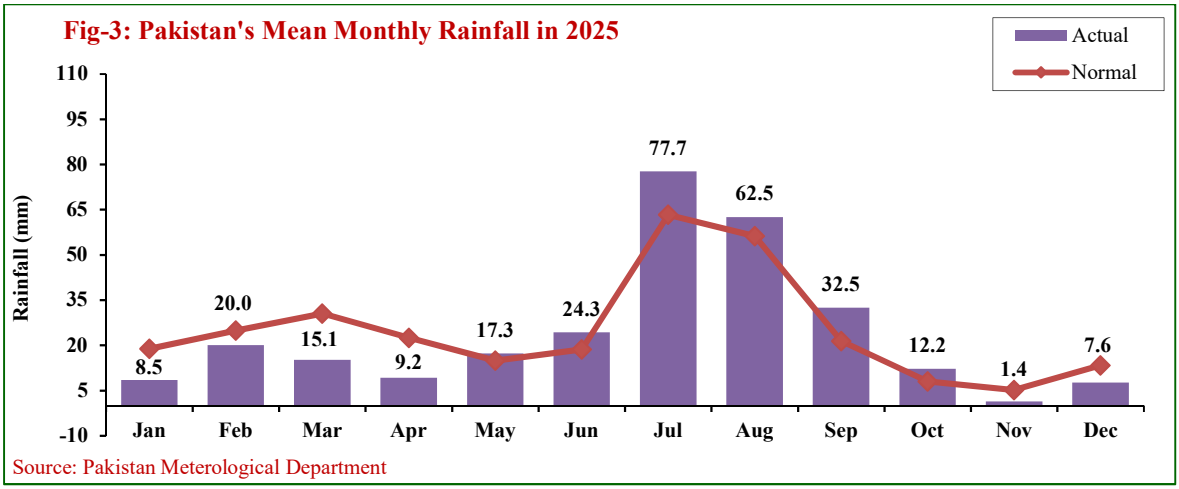
[12]State of Pakistan Climate Report 2025, Pakistan Metrological Department (Pakistan_Climate_2025.pdf)



17.2-b (ii) Rainfall: Below-Average Annual Rain

For the year 2025, the national total rainfall appeared at 288.5 mm (with near average of 297.6 mm (1961-2010)) with slight negative departure of -3 percent. Khyber Pakhtunkhwa with 543.4 mm (-21 percent) ranked 8th Driest (least rainfall was 502.0 mm in 1985) and Balochistan with 125.2 mm (-22 percent) observed below average. AJK with 879.3 mm (-6 percent) was near average. It was above average over Sindh with 230.1 mm (positive anomaly of 34 percent), GB with 232.2 mm (20 percent higher) and Punjab with 434.6 mm (12 percent higher).

Fig-3 represents synoptic mean monthly rainfall trend at country level. The year 2025 started with an excessively below-average rainfall. The national quarterly (Jan-Mar, JFM 2025) rain with 43.7 mm (-41.2 percent) recorded the 9th driest well below average rainfall. Similarly, the Pre-Monsoon (April-June, AMJ 2025) came up with -9 percent below average rainfall. However, monsoon season (July–September, JAS 2025) aggregate rainfall proved to be above 23 percent higher average with national total rainfall of 172.8 mm. Lastly, the post-monsoon season (October to December, OND 2025) appeared with the aggregate rainfall of 21.5 mm (-19 percent), was below the average across the country.



With regards to air pollution, as per 2025 IQAir World Air Quality Report[13],Pakistan ranked as the most polluted country in the world in 2025 (despite a reduction of 6.6 µg/m³ in average

[13]World Air Quality Report, 2025, IQAir (2025 World Air Quality Report | IQAir)

PM2.5 levels compared to the last year), with a population-weighted annual average PM2.5 concentration reaching 67.3 µg/m³ (thirteen times higher than the WHO guideline). Further there is evidence of concentration of urban hotspots, as the country appeared home to four of the world's ten most polluted cities, with Faisalabad recording the highest national average at 98.8 µg/m³. Broadly, 13 cities of the country reported PM2.5 levels exceeding 50 µg/m³. At the same time, seasonal variations further intensified pollution levels, with PM2.5 concentrations rising during April and May due to dust storms, and in Karachi during November where levels increased by nearly 57 percent compared to 2024. The crisis is primarily driven by several sources, which include industrial emissions, vehicle exhaust, brick kilns and construction dust. Thus, air pollution has continued to be a significant health issue affecting the population, reportedly causing almost 22,000 deaths every year and also straining ecosystems.

17.3 FY 2026 Government Performance Overview: Institutional Arrangements and Key Developments

Ministry of Climate Change and Environmental Coordination (MoCC&EC) has advanced Pakistan's climate resilience, adaptation, mitigation, and green development agenda during July–March 2026 through a range of strategic initiatives, policy reforms, international engagements and climate finance interventions. The Ministry prioritized strengthening institutional preparedness for climate induced disasters; enhancing glacier conservation and ecosystem restoration, promoting climate resilient infrastructure and localized adaptation planning, and mobilizing green and innovative financing mechanisms aligned with national priorities and international commitments under the Paris Agreement and the Sustainable Development Goals (SDGs).

Notable progress was achieved in operationalizing carbon market mechanisms, advancing the Pakistan Green Taxonomy, leveraging international climate finance, and fostering green skills, entrepreneurship, and sustainable economic transformation. Collectively, these efforts underscore Pakistan's commitment to transitioning toward a resilient, low-carbon, and environmentally sustainable development pathway while safeguarding vulnerable communities and ecosystems from the escalating impacts of climate change.

17.3-a Pakistan at 30th session of the Conference of the Parties (COP30)

The 30th Conference of the Parties (COP30) to the UNFCCC, held in Belém, Brazil, from 10th-22nd November 2025. Framed as the "Implementation COP" under the theme "Mutirão" (collective effort), the conference focused on translating climate commitments into actionable outcomes and strengthening multilateral cooperation for effective implementation of the Paris Agreement. The details of major outcomes and Pakistan's participation in COP30 are presented in Box-I.

Box-I: COP30 Agenda and Pakistan's Strategic Role

COP30 adopted the Belém Political Package, anchored in the Global Mutirão decision, with several outcomes of direct relevance to Pakistan. These included the launch of a Global Implementation Accelerator to support implementation of Nationally Determined Contributions (NDCs) and National Adaptation Plans (NAPs); a strong emphasis on climate finance, including a call to triple adaptation finance by 2035, establishment of a two-year work programme on finance under Article 9.1, and enhanced ex-ante finance reporting (Article 9.5) to improve predictability and transparency and operationalization of the Loss and Damage Fund under the Barbados Implementation Modalities with an initial allocation of US $ 250 million, based on grant financing and simplified access. Progress was also achieved on the Global Goal on Adaptation (GGA) through adoption of 59 Belém Adaptation Indicators across key sectors, designed to remain voluntary and country-driven, reflecting Pakistan's position on flexibility. Additional outcomes included establishment of a Just Transition Mechanism emphasizing nationally determined pathways and support for developing countries in operationalization of the Technology Implementation Programme (TIP) to accelerate technology transfer and innovation.

Pakistan positioned itself as a leading voice among developing countries, advocating for adaptation, climate resilience, equity and scaled-up means of implementation. At the Pakistan Pavilion, MoCC&EC organized a series of high-level side events on key themes included cryosphere adaptation and disaster risk reduction in the Hindu Kush–Karakoram–Himalaya region, implementation of Pakistan's NDC 3.0, operationalization of loss and damage mechanisms, gender dimensions of heatwaves and energy poverty, transformation of climate finance architecture, promotion of green skills and climate innovation. It also highlighted scaling up of the Green Pakistan Programme which has achieved plantation of over 2.25 billion trees and restoration of approximately 700,000 hectares. COP30 also witnessed extensive participation from provincial governments and national institutions. Provinces showcased initiatives on climate-resilient urban planning, flood recovery housing, forestry interventions and green growth. The Punjab delegation highlighted major initiative including the international launch of "Suthra Punjab", a large-scale integrated solid waste management system demonstrating innovation in circular economy practices.

Source: Ministry of Climate Change and Environment Coordination

17.3-b Climate Resilience and Adaptation Initiatives

i. Monsoon 2026 Strategic Plan

In the wake of the devastating 2025 monsoon floods and its pronounced impacts, the Monsoon 2026 Plan has been sets forth as a proactive framework to mitigate risks, strengthen resilience and safeguard communities against future climate-driven disasters. Pakistan's Monsoon 2026 Strategic Plan, prepared by MoCC&EC and approved by the Honourable Prime Minister in November 2025, constitutes a time-bound and anticipatory climate adaptation framework. Acknowledging the country's widespread exposure to riverine, urban, hill-torrent, glacial lake outburst flood (GLOF), and coastal flooding, the Plan adopts a risk-informed approach encompassing targeted structural, non-structural and institutional interventions.

The 240-day preparedness phase emphasizes rapid rehabilitation of embankments and critical flood infrastructure, completion of small dams, restoration of urban drainage systems, enforcement of floodplain zoning regulations, and integration of real-time telemetry with forecasting systems and reservoir operations. Community-level resilience is further enhanced through the provision of mobile health and WASH services, establishment of temporary learning facilities and improved access to district-level early warning systems. The strategic plan measures are complemented by investments in advanced meteorological monitoring and the development of an integrated national early warning system, supported by governance reforms to ensure coordinated water and climate management. Collectively, the Plan seeks to transition Pakistan from a predominantly reactive response paradigm to a proactive, resilience-oriented flood management framework, thereby safeguarding lives, livelihoods and critical infrastructure from intensifying monsoon-related risks. Implementation is being carried out in close coordination with line ministries, relevant departments and provincial governments, including AJK and GB.

ii. Glaciers Preservation and Resilience Initiatives

Pakistan hosts more than 13,000 glaciers, forming a significant component of Hindu Kush Karakoram Himalaya, often referred to as the "Third Pole", which are vital for national water security, agriculture, hydropower generation and local livelihoods. Despite Pakistan's relatively low contribution to global greenhouse gas emissions, rising temperatures and the deposition of light absorbing particulates such as black carbon, are accelerating glacier retreat, thereby increasing the risk of hydrological instability and downstream hazards. In response, the government has prioritized resilience-building and disaster risk reduction interventions. The completion of the GLOF-II project in December 2025 marked a significant milestone, enhancing early warning systems, strengthening institutional capacity and improving community-level preparedness in vulnerable glacial regions. Recognizing the transboundary and global significance of cryospheric change, Pakistan presented a cross-regional Glacier Preservation and Resilience

Initiative at COP30 under the United Nations Framework Convention on Climate Change framework, where it received broad international support.

Building on this momentum, Pakistan has initiated the planning phase for GLOF-III and is concurrently developing a comprehensive national Glacier Preservation and Resilience Strategy. These efforts underscore the strategic prioritization of glacier systems within the country's climate policy framework, reflecting their critical importance for long-term environment sustainability and socio-economic stability.

iii. Integrated Climate Risk Management in Buner and Shangla Districts of Khyber Pakhtunkhwa

The Buner and Shangla Climate Adaptation Project has been initiated in March 2025 in Khyber Pakhtunkhwa, with a total allocation of US $ 9.8 million, financed by the Green Climate Fund. The project focuses on strengthening early warning systems, enhancing local institutional capacity and promoting community preparedness through structured training and awareness programmes. It is expected to benefit over 1.6 million people, enabling vulnerable populations in these mountainous districts to better withstand the escalating impacts of climate change.

iv. Sustainable Actions for Ecosystem Restoration in Pakistan (SAFER Pakistan Project)

A project titled "SAFER Pakistan Project" launched in September 2025 is a major climate adaptation initiative funded by the Adaptation Fund and led by Pakistan's MoCC&EC in partnership with the International Centre for Integrated Mountain Development (ICIMOD). With a portfolio of US $ 10 million, it aims to strengthen resilience against climate change by focusing on water security. The project focuses on nature-based solutions and integrated water resource management, including community-based early warning systems, spring and groundwater recharge, climate-resilient water infrastructure, wetland restoration and wastewater treatment and rehabilitating traditional water bodies. It also emphasizes social inclusion by building the capacity and participation of women, youth and children in adaptation planning and disaster resilience.

v. National Adaptation Plan Progress: DAPs

Under National Adaptation Plan, the preparation of District Adaptation Plans progress. This initiative is a climate adaptation effort aimed at strengthening local resilience by developing inclusive, evidence-based strategies for addressing climate risks at the district level. It focuses on building the capacity of district governments and communities to assess climate vulnerabilities, prioritize adaptation actions, and integrate climate-resilient planning into local development processes. Through stakeholder engagement, data collection, risk analysis and training, the initiative helps districts design and implements tailored adaptation measures across key sectors such as water, agriculture, disaster risk reduction and infrastructure. District adaptation plans for Badin and Dera Ghazi Khan have already been finalized, while plans for Karak and Charsadda are currently in process.

vi. Climate and Disaster Resilience Enhancement Programme (CDREP) – Subprogramme II:

For strengthens institutional and financial preparedness for disasters, the Government of Pakistan, through the Finance Division as the Executing Agency, is implementing Asian Development Bank (ADB)-supported Climate and Disaster Resilience Enhancement Programme (CDREP) with a total programme size of US $ 1.0 billion, comprising Subprogramme-I and Subprogramme-II. CDREP adopts an integrated policy reform approach focusing on: (i) improving institutional capacity for disaster planning, preparedness and response (ii) increasing investments in disaster risk reduction (DRR) and climate resilience and (iii) operationalizing a risk-layered Disaster Risk Financing (DRF) framework to ensure timely availability of financial resources during disaster events.

Subprogramme-II (September 2024 – March 2026)

Subprogramme-II (SP-II), with a financing envelope of US $ 500 million (US $ 250 million

OCR and US $ 250 million concessional OCR), is currently under implementation. The loan carries a 15-year term, including a 3-year grace period, with an interest rate of SOFR + 0.5 percent. A key feature of SP-II is the Contingent Disaster Fund (CDF) of up to US $ 500 million, designed as a multi-year facility to provide immediate liquidity in the event of disasters, with disbursements routed through the national budget. Under SP-II, 24 policy actions across 11 government entities are being implemented, expected to be completed by December 2025. The key reforms include:

- Adoption of the National Disaster Management Plan III (2024)
- Development of a Natural Catastrophe (NatCat) Risk Model
- Introduction of Gender-Responsive Disaster Management Frameworks
- Establishment of Blended Finance Frameworks and Climate Screening Mechanisms

The programme also promotes innovative climate finance instruments, including green sukuk and supports the development of a National Climate Finance Strategy. The DRF Strategy aims to enable rapid financial response and support vulnerable populations through mechanisms such as the Benazir Income Support Programme (BISP). ADB is providing technical assistance (US $ 1 million) to strengthen implementation capacity, focusing on institutional strengthening, development of innovative financing instruments (insurance, catastrophe bonds) and programme monitoring. Overall, CDREP SP-II represents a critical step toward enhancing Pakistan's institutional, financial and operational capacity to manage climate-induced disasters and safeguard sustainable development.

17.3-c Capacity Building on Water Quality Monitoring SDG 6 (6.1) Reporting

The capacity building regarding reporting of water quality monitoring initiative is a climate adaptation and sustainable development intervention designed to strengthen Pakistan's ability to track and report progress toward Sustainable Development Goal 6.1 with a portfolio of Rs 102.0 million. The initiative is critical as the SDG 6.1 aims to achieve universal and equitable access to safe and affordable drinking water. By improving institutional systems for consistent, accurate and transparent water quality monitoring, the initiative supports evidence-based decision-making, compliance with national and international standards and integrate water quality data into planning and policy frameworks.

17.3-d Submission of Nationally Determined Contributions (NDC 3.0)

In fulfillment of commitments under the United Nations Framework Convention on Climate Change and the Paris Agreement, Pakistan submitted its third Nationally Determined Contribution (NDC 3.0) on 24th September 2025. This submission articulates an integrated and forward-looking framework that embeds both mitigation and adaptation priorities within national development planning. It outlines cross-sectoral interventions spanning energy, industry, agriculture, forestry, waste, water, health, education, housing and social protection systems, with the objective of advancing climate resilience, sustainability and inclusive development. Pakistan remains committed to reduce projected greenhouse gas emissions by up to 50 percent relative to a business-as-usual trajectory. Under NDC 3.0, this target is extended to 2035, with 17 percent to be achieved unconditionally through domestic policy measures and resource mobilization and an additional 33 percent contingent upon the provision of adequate (primarily grant-based or concessional) international climate finance, alongside technology transfer and capacity-building support.

However, progress has been constrained by limited access to the required international financing. The total investment needed to achieve these targets is estimated at US $ 565.7 billion. The World Bank, in its Pakistan Country Climate and Development Report (2022), estimated baseline climate-resilient and low-carbon investment needs at US $ 348 billion up to 2030, implying an additional requirement of approximately US $ 217.7 billion by 2035.

Overall, NDC 3.0 reflects a dual strategic orientation: advancing low-carbon development while strengthening resilience to climate-induced shocks. It underscores that the scale of ambition exceeds domestic capacities alone, thereby necessitating sustained international cooperation, technology partnerships and financial support to enable effective implementation while safeguarding long-term development gains.

17.3-e Submission of Third National Communication of Pakistan

Pakistan submitted its Third National Communication (TNC) to the United Nations Framework Convention on Climate Change on 30th June 2025, representing a critical milestone in the country's climate reporting and policy framework. A central component of the TNC is the national greenhouse gas (GHG) inventory for 2020–21, prepared by the Global Climate-Change Impact Studies Centre (GCISC). The inventory was developed in accordance with the IPCC 2006 Guidelines for National Greenhouse Gas Inventories, utilizing standardized tools and methodologies, including the IPCC GHGI Software, to ensure transparency, consistency and comparability. Beyond fulfilling reporting obligations, the TNC serves as a strategic, evidence-based foundation to guide Pakistan's transition toward a low-carbon and climate-resilient development pathway. MoCC&EC as the national focal institution for climate governance has led this process with technical support from GCISC.

17.3-f Pakistan Climate Change Authority (PCCA)

In response to the climate burdens, the Government of Pakistan has established Pakistan Climate Change Council, Pakistan Climate Change Authority (PCCA) and Pakistan Climate Change Fund (PCCF) under the Pakistan Climate Change Act 2017 as the key milestones to strengthen climate governance in the country. As a newly established national technical entity, the PCCA is mandated to mobilize international climate finance and support coordinated implementation and monitoring of national climate policies, strategies and programmes in collaboration with the provinces, in line with international commitments and to pursue low emission and climate-resilient development pathways.

During 2025-26, the institutional set up, operational rules and procedures for effective function of PCCA and for the establishment of PCCF were formulated as the essential foundational steps. So, far the authority has played an effective role in accessing climate by formulation of more than 38 project proposals and successfully secured three projects. Currently, PCCA is in process of engaging with GCF as the first-ever national entity for Direct Access Readiness Support from GCF in collaboration with MoCC&EC. The proposed readiness support aims to establish a mechanism for coordinated implementation, monitoring, data management and reporting of climate interventions.

17.3-g Establishment of Green University and National Green Startup Innovation Challenge (NGSIC)

For promotion of green skills in Pakistan, MoCC&EC is advancing the establishment of a "Green University in Pakistan" as a strategic initiative to develop a climate resilient, future-ready human capital base. The initiative aims to equip young professionals with advanced green skills while fostering innovation, green entrepreneurship and applied research in climate and environmental domains. Strategically, the university is envisioned to produce a skilled and adaptive workforce capable of driving Pakistan's transition toward a low-carbon, resource-efficient and climate-resilient economy.

Further, MOCC&EC is organizing a "Shark Tank" style Green Startup Innovation Challenge which will create a high-visibility, competitive, and investment-focused platform. The initiative will promote climate-smart entrepreneurship in Pakistan by identifying, supporting, and scaling innovative green startup solutions with high environmental, social, and economic impact. It will help in establishing a national-level investment pitching platform where green startups and innovators present their business ideas to a panel of investors for funding,

mentorship and partnership opportunities. The intended outcomes of the event are to enhance number of viable green startups, enhanced youth engagement in climate action and green economic growth. It will contribute towards Pakistan's NDC targets, green industrial policy and national green jobs agenda.

17.3-h Initiatives Related to Ecosystem, Forestry and Biodiversity

Pakistan is advancing integrated long-term interventions focused on ecosystem restoration and climate adaptation, including forest landscape restoration, mangrove rehabilitation and blue carbon initiatives in the Indus Delta. At the international level, Pakistan is an active member of the Riyadh Global Drought Resilience Partnership and ratified the Middle East Green Initiative Charter in 2025, strengthening regional collaboration on drought resilience, land restoration and nature-based solutions. The country continues to operationalize Land Degradation Neutrality targets through integration into national planning processes.

Concerning forestry, Pakistan's total forest area is estimated at approximately 4.1 million hectares, representing about 4.7 percent of the national land area. The average annual deforestation rate is estimated at around 11,000 hectares, reflecting persistent pressures on forest resources. MoCC&EC, in collaboration with provincial and territorial governments is implementing large-scale afforestation and ecosystem restoration initiatives to enhance forest cover. Under the Green Pakistan Programme, with a total allocation of Rs 122.1 billion, approximately 2.3 billion plants have been raised, regenerated and distributed nationwide. By March 2026, a cumulative total of 2,283.3 million plants had been planted, regenerated, or distributed, achieving 69.3 percent progress and increasing tree cover over about 0.7 million hectares. As reported by the provinces /territories, a total of 30.5 million plants were planted, regenerated or distributed during July–March 2026. The province-wise and special areas performance against plantation targets remained as under: Punjab (79 percent), Sindh (86 percent), Khyber Pakhtunkhwa (73 percent), Balochistan (28 percent), AJK (34 percent) and GB (62 percent).

In the biodiversity sector, Pakistan is revising its National Biodiversity Strategy and Action Plan in alignment with the Kunming-Montreal Global Biodiversity Framework. The country has also ratified the Agreement on Biodiversity Beyond National Jurisdiction, aimed at strengthening global governance for marine biodiversity conservation. Efforts to expand protected areas continue, with a total of 476 protected areas now covering over 20 percent of the country's territory, including marine ecosystems.

17.3-i Key Development Related to International Cooperation

Pakistan ratified Kigali Amendment in 2025 via presidential signing. The Kigali Amendment to the Montreal Protocol is a legally binding international agreement aimed at phasing down hydrofluorocarbons (HFCs) and potent greenhouse gases. Its significance lies in reducing global warming by up to 0.5 °C by 2100, promoting energy-efficient and eco-friendly cooling technologies. Furthermore, an import licensing system had been introduced for the import of hydrofluorocarbons (HFCs) which was a mandatory requirement of the Kigali Agreement. Additionally, Pakistan has also imposed a ban on the import of chloroform.

17.4 Pakistan Climate Finance: Performance Insights of FY 2026

Apart for advancing climate resilience and adaptation through institutional arrangements and key integrated policy developments, the Government of Pakistan has also made following significant progress in regards to climate finance and investment frameworks.

17.4-a Pakistan Climate Prosperity Plan (CPP) and Investor Framework

Pakistan Climate Prosperity Plan (CPP) is the Government's flagship investment-led framework designed to transform climate challenges into opportunities for sustainable

economic growth, resilience and prosperity. It serves as a comprehensive climate investment blueprint aligned with Pakistan's NDCs, national development priorities and global sustainability commitments. The CPP has been launched by the Finance Division on 29th April 2026, aiming to reposition climate action as a driver of economic transformation by mobilizing public, private and international finance and by embedding climate resilience across key sectors of the economy.

Strategic Vision and Investment Framework of CPP

The CPP targets a transition toward a climate-resilient and low-carbon economy through: 1) Expansion of renewable energy and clean technologies 2) Promotion of climate-resilient agriculture and food security 3) Development of green industrial zones and circular economy models 4) Strengthening of climate-resilient infrastructure, 5) Scaling up financial protection mechanisms and disaster risk financing and 6) Protection and restoration of natural capital through nature-based solutions.

The plan estimates an overall investment requirement of approximately US $ 1.6 trillion by 2050, with annual investment needs of around US $ 65 billion. Investments are projected to reach US $ 565.7 billion by 2035. To operationalize implementation, a Country Platform led by the Finance Division in coordination with key ministries and provinces will facilitate project pipeline development, mobilize financing and strengthen institutional coordination. In this regard, the CPP identifies priority sectors including energy transition, agriculture, green economic zones, infrastructure, transport and natural capital. The key targets include:

- Transition to 60 percent clean energy by 2030 and 95 percent by 2040
- Achieving 30 percent EV penetration by 2030
- Boosting agricultural productivity by 30 percent through climate-smart practices
- Climate-proofing 75 percent of infrastructure by 2040
- Expanding protected natural areas and restoring degraded ecosystems

In regards, to Investment Pipeline and Investor Engagement, there is CPP Investor Guide, which presents a pipeline of climate-aligned, bankable projects across sectors including renewable energy, hydropower, electric mobility, climate-smart agriculture, waste management and carbon markets. In particular, these projects include: 1) Hydropower and solar energy projects 2) Electric vehicle manufacturing and mobility solutions 3) Climate-smart agriculture and biochar initiatives 4) Wastewater treatment and circular economy projects and 5) Carbon offset and afforestation initiatives. The investor framework emphasizes blended finance, public-private partnerships (PPPs) and innovative instruments such as green bonds, sukuk, and carbon markets to attract private capital and reduce financing constraints.

Regarding macroeconomic and development impact, the CPP is expected to promote green growth and reduce climate-induced economic losses, enhance energy security and reduce import dependence, strengthen fiscal resilience through reduced disaster-related expenditures, generate employment through green sectors and lastly, improve environmental sustainability and public health outcomes.

17.4-b Pakistan Green Taxonomy (PGT)

Pakistan Green Taxonomy has been approved and notified on 15th September, 2025. It is a classification system for sustainable economic activities, developed to guide investment towards environmental objectives such as climate change mitigation and adaptation. It categorizes activities to prevent green washing by providing a standardized definition of green projects, making it easier to mobilize capital for sustainable development. The implementation is on fast-track in collaboration with Securities and Exchange Commission of Pakistan (SECP) and State Bank of Pakistan (SBP).

The SECP has implemented the green taxonomy by issuing revised ESG Disclosure Guidelines for listed companies, aligning corporate reporting with PGT. This framework mandates reporting of climate related risks, opportunities, and for large listed companies, taxonomy-aligned activity data, starting voluntarily until June 2029 before a three-phase mandatory rollout. The primary goals are to increase investor confidence, improve the credibility of climate disclosures and direct capital towards sustainable, green projects in line with national climate goals.

17.4-c Carbon Markets

Pakistan policy guidelines for Trading in Carbon Markets have been operationalized, incentivizing mitigation activities under the market-based mechanisms of Article 6 of the Paris Agreement. Under the policy guidelines, four projects have been issued Letters of Intent during the period July–March 2026. The projects details include:

i. Biomethane, Bio-Pellets & Organic Fertilizer Project

The project began in October 2024 with a 15-year crediting period. It is expected to achieve emission reductions of 12,299,429 tCO2e over 25 years' operational lifetime, averaging 491,977 tCO2e annually. The project will convert cow dung, fruit and vegetable waste into biogas and organic fertilizer. As an outcome, the principal product is biogas, a clean-burning fuel derived from organic waste, which will also be converted into bio-methane gas.

ii. Lakhodair Landfill Site Methane Capturing Project

The project has an estimated credit of 225,000 tCO_2e per year over a 10-year crediting period, with the first issuance expected in 2027. It will reduce greenhouse gas emissions and air pollution by capturing methane and preventing open burning of waste. The project will result in solid waste management and protection of public health from the risks posed by unsafe disposal of waste material.

iii. Pakistan Finance for Lowering Emissions in Waste and Water Sectors Project in Punjab

The project is estimated to generate 1,184,791 tCO_2e credits between 2026 and 2030. Its outcomes include wastewater treatment, clean drinking water supply and the use of renewable energy in operations.

iv. Pakistan Finance for Lowering Emissions in Waste and Water Sectors Project in Sindh

The project has an estimated credit of around 0.5 million tCO_2e per year. Its outcome focuses on improved solid waste management in the Sindh province.

17.4-d Development of Climate Finance Dashboard

Climate Finance Dashboard is under development to ensure transparency and accountability in the inflow of funds mobilized nationally and internationally under the climate finance. The dashboard will have real time data related to all kinds of climate budget tagged finance mobilized nationally through PSDP, ADPs, Private Sector Investments and internationally donor-funded projects all over the country. The purpose is to digitize information based on validated data which will eventually help draft strategies related to activities under adaptation, mitigation and other supporting areas. It will also ensure completeness and authenticity of future climate finance data to be collected through well-defined protocols from federal and provincial stakeholders.

17.4-e IMF Resilience and Sustainability Facility (RSF) – Reform Progress

Pakistan is implementing a comprehensive set of climate-related reforms under the IMF's Resilience and Sustainability Facility (RSF), aimed at strengthening climate resilience, promoting green growth and improving fiscal sustainability. The progress update in this regard is presented in Box-II.

Box-II: IMF's RSF Progress Update

The key "Completed Reforms" under the programme include:

- Introduction of a supplementary carbon levy on petroleum products through the FY 2026 Finance Act.
- Adoption of a revenue-neutral electric vehicle (EV) policy framework, including subsidies for EVs and taxation of internal combustion vehicles.
- Issuance by State Bank of Pakistan (SBP) of climate-related financial risk management and stress testing guidelines.
- Operationalization of green taxonomy and ESG disclosure guidelines by Securities and Exchange Commission of Pakistan (SECP), with phased mandatory reporting from 2029.

Whereas, the "Ongoing and Pipeline Reforms" comprise of:

- Integration of climate weighting (minimum 30 percent) in PSDP project selection and improved climate screening.
- Development of a National Disaster Risk Financing Strategy implementation framework.
- Reform of energy subsidies, transitioning to targeted support mechanisms for low-income households through BISP.
- Promotion of EV charging infrastructure through viability gap funding (VGF) frameworks.
- Introduction of irrigation water tariff reforms (e-Abiana) to improve cost recovery and water efficiency.
- Expansion of Minimum Energy Performance Standards (MEPS) and energy-efficient public procurement.
- Strengthening of climate budget tagging and reporting, including quarterly execution reports.

These reforms collectively aim to reduce carbon intensity, enhance climate resilience, improve resource allocation and mobilize sustainable finance, while addressing structural inefficiencies in energy, water and public investment systems.

Source: Ministry of Finance

17.4-f Glaciers to Farms (G2F) Regional Programme: Advancing Climate Resilience & Sustainable Development in Central and West Asia

In November 2025, a Green Climate Fund Regional Programme Glaciers to Farms (G2F) has been approved and Pakistan has been allocated US $ 27.5 million against which concepts notes are being developed and sifted to pitch projects.

17.4-g Green Climate Fund (GCF) Financing

The "Artificial Recharge Interventions for Ground Water & Climate Adaptive Watershed Management in Quetta Sub Basin" project proposal received from the Government of Balochistan was approved by the Pakistan GCF Board on 17th December, 2025 and later forwarded to GCF international for approval through NRSP being accredited agency. The project proposes an integrated solution to Quetta's water crisis by constructing a network of 205 recharge wells and 113 injection wells strategically located in permeable zones near existing check dams and natural flood channels. These are synergistically combined with the afforestation of 350 acres of degraded watershed using native, drought-resistant species and the installation of a real-time monitoring network of 50 sensor-equipped wells.

The project will cost US $ 9.0 million and indicative funding from GCF would be around US $ 5.6 million. It will benefit 2.5 million people directly. Estimated project lifespan is 20 years with the implementation period of 3 years. The project will contribute to reduce 20 percent emissions and 80 percent in adaptation with increased resilience and livelihood of the

communities based in Quetta. Approximately 66,000 to 67,000 tCO2e sequestration expected in 20 years (141.6 hectares afforestation planned) under the project. The artificial recharge interventions and watershed restoration measures through afforestation are expected to remain effective and continue generating climate adaptation benefits beyond the project completion.

17.4-h Global Biodiversity Framework Fund (GBFF) Financing

Under the 2nd Tranche of Global Biodiversity Framework Fund (GBFF) with indicative ceiling for Pakistan US $ 2.8 million, MoCC&EC has endorsed two projects (i) Nature-based Solutions (Nbs) in form of Wetlands and reforestation for Biodiversity Enhancement and Climate Change Adaptation in the Riverine and Marine Urban Context-Lyari River, Karachi with proposed GEF financing of US $ 1.9 million and (ii) Accessing the benefits of Biodiversity: Working towards the national Legislation and Implementation of the Nagoya Protocol with proposed GEF financing of US $ 1 million). Upon approval from GBFF, Pakistan would leverage US $ 2.8 million as climate finance.

17.5 Natural Disasters of 2025–2026 in Pakistan

The overview of natural disasters appeared in Pakistan during 2025–2026 and their associated human and economic impacts are detailed below.

Floods of 2025

Pakistan, consistently ranked among the most climate-vulnerable countries, is recurrently exposed to climate-induced hazards due to its location within the Indus River Basin, the intensity of the South Asian monsoon system and the cryospheric dynamics of the Hindu Kush–Himalayan region. These wide spectrums of interlinked climate-induced and hydro-climatic hazards include floods, droughts, heatwaves, GLOFs, cloudbursts, extreme monsoon precipitation, riverine flooding and flash floods and severe smog episodes. The risks from hydro-climatic hazards are amplified by the highly integrated nature of the Indus Basin hydrological system, where major rivers such as the Indus, Jhelum, Chenab, Ravi and Sutlej River can experience concurrent peak flows during extreme rainfall events, overwhelming downstream infrastructure and flood protection systems.

The frequency and intensity of these events have risen markedly in recent years, reflecting escalating climate stress. During the 2025 monsoon season, rainfall beginning in late June rapidly evolved into a large-scale compound disaster. According to Pakistan Metrological Department, monsoon season (July–September, JAS 2025) aggregate rainfall proved to be above average (positive anomaly of 23 percent) with national total rainfall of 172.8 mm. Overall, all provinces and territories have experience above average rainfall with Punjab with 310.1 mm (34 percent higher), GB with 51.9 mm (31 percent higher), Balochistan with 74.8 mm (28 percent higher), Sindh with 168.8 mm (26 percent higher) all exhibited above average rainfall, while AJK with 407.8 mm (5 percent higher) and Khyber Pakhtunkhwa with 260.7 mm (2 percent higher) saw near average monsoon rains. This heightened precipitation, combined with enhanced glacier melt, elevated river discharges, inadequate drainage capacity and unsustainable land-use practices, led to widespread inundation across Khyber Pakhtunkhwa, GB, AJK and Punjab. The interaction of these drivers transformed a seasonal monsoon into a multi-hazard disaster. The situation further intensified in late August 2025 when Sutlej, Ravi and Chenab River experienced synchronized flood peaks, producing unprecedented compound flooding across Punjab. Thousands of people were displaced and villages were inundated, causing substantial agricultural losses with implications for national food security. Damage to embankments, irrigation networks and rural infrastructure compounded humanitarian needs and magnified economic losses. The details of these damages in economic terms are reported in Box-III.

Box- III: Economic and Sectoral Impacts of 2025 Floods

The 2025 flooding caused widespread economic and humanitarian devastation, with total damages estimated at Rs 822 billion (US $ 2.9 billion). The disaster resulted in over 1,039 fatalities, displaced more than 4 million people and affected approximately 6.5 million individuals across 70 districts, with nearly 77 percent of losses concentrated in Punjab.

According to the Flood Assessment Report 2025 (data as of 30th September 2025), the agriculture sector incurred damages of Rs 430 billion, including Rs 422.6 billion in crop losses, while infrastructure-related damages were estimated at Rs 307 billion. Within overall infrastructure losses of Rs 307 billion, major damage includes of road networks accounted Rs 187.7 billion, followed by housing infrastructure at Rs 91.2 billion, whereas other sectors, including bridges, water and energy infrastructure, sustained combined damages of Rs 28.05 billion. Regionally, the distribution of damages is highly uneven, heavily concentrated in Punjab accounting Rs 631 billion (76.8 percent) of total losses.

Sectoral impacts were particularly pronounced across the agriculture and infrastructure sectors. In agriculture, critical crops including cotton and rice sustained damages, thereby undermined food security and export potential. At the same time, widespread destruction of physical infrastructure, including thousands of kilometers of road networks, bridges and more than 229,763 housing units, has imposed severe strain on public resources and recovery capacities.

These floods disruptions are also projected to have labor market implications, with unemployment expected to increase by approximately 0.2 million in FY 2026. Consequently, the floods have emerged as a significant downside risk to Pakistan's macroeconomic outlook, with real GDP growth for FY 2026 revised downward to an estimated range of 3.5-3.9 percent compared to the initial target of 4.2 percent.

Source: Ministry of Climate Change and Environment Coordination

The latest update as of 5th December 2025 by EM-DAT, the floods of 2025 have caused the estimated US $ 3.0 billion in damages, affected 6.9 million people and resulted in 1,037 deaths due to the monsoonal rains. Additionally, so far in 2026, despite only a few months into the year, the severe weather event in the form of storm, which lasted just one day in the Karachi region (Sindh province) has caused economic losses of US $ 5.0 million alone to the economy (Table 17.1).

Table 17.1: Natural Disasters of 2025-2026 and their Consequences in Terms of Human Life and Damages

Year	Disaster Type	Disaster Sub Type	Total Deaths	Total Affected	Total Damages ('000 US $)
2025	Storm	Storm (General)	10	21	-
	Flood	Flash flood	1,037	6,901,067	3,000,000
	Storm	Severe weather	20	150	-
2026	Earthquake	Ground movement	1	152	-
	Storm	Severe weather	83	2,315	-
	Storm	Severe weather	19	50	5,000
Total			1,170	6,903,755	3,005,000

Source: EM-DAT (The International Disaster Database)

17.6 Provincial Performance Regarding Environment Protection: Achievements and Key Initiatives of FY 2026

Punjab

- Environment and Climate Change Endowment Fund has been established on 20th September 2025 with the capitalization of Rs 6,700 million under the World Bank-assisted Punjab Green Development Programme (PGDP), in line with the Punjab Green Financing Strategy, to achieve Disbursement Linked Result (DL.R 8-1).

The fund serves as a sustainable financing mechanism, operationalized through the not-for-profit Punjab Environment & Climate Change Endowment Fund (PECCEF) Company to promote green investments, environmental protection, emissions reduction and enhance climate resilience.

- Similarly, a Green Climate Fund (GCF) Accreditation Unit has been constituted on 10th October 2025, by the Environment Protection Climate Change Department. The unit serves as the focal entity for all matter pertaining to GCF accreditation, ensuring coordination with MoCC&EC and compliance with fiduciary and institutional requirements in accordance with the GCF framework.
- The CM Punjab Climate Internship Programme was approved on 22nd October 2025 as a provincial initiative to strengthen climate governance through structured youth engagement. The programme bridges the gap between academic leaning and field implementation by recruiting 1,250 interns from diverse academic backgrounds and all districts of Punjab.
- Punjab Green School Certification Programme has been formally initiated with Phase-I launched on 12th September 2025. The programme promotes school-level climate action across 41 districts through six thematic sectors (i) Air for Life (ii) Every Drop Counts (iii) Save Energy (iv) Waste Wise (v) Environmental Action and (vi) Plant for Pakistan. Till end December 2025, 16,461 schools have pledged participation, including 14,934 public schools (91 percent) and 1,527 private schools (9 percent), reflecting strong institutional uptake of the programme across the province.
- Further, with regards to digitalization, Punjab Environment Protection Agency (EPA) has introduced wide range of initiatives, mainly include:

i. **Industrial Directory, E-Mapping, AI-based Monitoring and System Integration:** EPA Punjab maintains a comprehensive industrial directory with assigned Unique Identification (ID) numbers to each industrial unit. It has also completed province-wide e-mapping of industrial units, enabling real-time tracking of location, operations and environmental compliance, further supported by CCTV-based monitoring with AI and machine learning for violation alerts. It also integrates the Planning Support System (PSS) with the Environmental Assets Management System (EAMS), supported by GIS-based mouza-level mapping, therefore improves policy planning and removes unnecessary procedural steps.

- Additionally, regarding institutional reform initiatives, EPA Punjab undertook a series of measures in 2025, to strengthen environmental governance and enforcement across the province. The details include:

i. **ISO 9001 Certification:** EPA Punjab achieved ISO 9001 certification in 2025 for its EIA process, becoming the first provincial environmental agency in Pakistan to meet this international quality management standard.

ii. **Air Quality Monitoring System (AQMS):** The EPA has strengthened province-wide Air Quality Monitoring Stations through installation of 46 monitoring stations across 18 districts. These stations conduct continuous, real time monitoring of local air quality and environmental parameters.

iii. **Drought Management Initiative:** For the first time, EPA Punjab took a proactive role in drought management by initiating coordinated measures to assess, monitor and address drought impacts through evaluation of environmental stress indicators, water

scarcity trends and ecosystem vulnerability to support informed planning and mitigation efforts.

- Further, during 2025 EPA Punjab conducted large-scale enforcement actions, including over 150,000 inspections in industrial and brick kiln sectors, resulting in demolitions, sealing actions, FIRs and fines exceeding Rs 687 million. Alongside significant interventions reported below, resulted into measurable environmental gains, including a 13 percent reduction in PM2.5 levels in Lahore as reported by IQAir 2025 report, expansion of the Air Quality Monitoring System from 3 to 51 stations with a target of 100 by April 2026 and enhanced real-time pollution monitoring across Punjab. The details of these initiatives are:

  i. **Hawk Eye Surveillance System:** EPA Punjab deployed the Hawk Eye vehicle with 360-degree live camera vehicle for real-time monitoring of environmental compliance. It is also supporting enforcement of regulations through daily surveillance of industrial hotspots, dust management and water recyclers installation monitoring and hotspots identification for fog cannon operations.

  ii. **Fog Cannon Fleet (Anti-Smog Initiative):** Pakistan's first ever fog cannon fleet has been launched, with 15 units operating in Lahore as part of air pollution mitigation strategy. The ultra-fine mist spray technology will effectively settle harmful PM10 particles, reducing level of suspended particles, thereby improving air quality locally.

  iii. **Brick Kiln Reforms (Zig-Zag Technology):** In 2025, EPA Punjab strengthened its enforcement against brick kiln ensured transition to zig-zag technology, achieving 100 percent compliance along motorway zones. Resultantly, the initiative substantially reduced particulate emissions and black carbon levels thereby improved air quality along major transport corridors.

  iv. **Anti-Stubble Burning Initiative:** EPA Punjab has launched a dedicated initiative to curb stubble burning, a major contributor to air pollution in the province. Some comprehensive initiatives in this regard include: Anti-Stubble Burning Squad, provision of 5,000 super seeders to farmers, deployment of drone and eight e-squad monitoring and establishment of 67 Quick Response Centers along major motorways (M-2, M-3, M-4 &M-11) in 8 Stubble Burning Hotspots Districts, which enabled prevention, monitoring and rapid response to crop burning incidents.

  v. **Construction Dust Control Measures:** The EPA has deployed modern mist sprinkler systems at construction sites as a part of anti-smog measure in Lahore as construction activities are significant contributors to dust and particulate pollution. These systems spray water over exposed soil, debris and construction materials, reducing the dispersion of airborne particulate matter thereby help maintain air quality.

  vi. **Thermal Night Vision Drone Surveillance:** The EPA for the very first time introduced thermal night vision drones to conduct aerial surveillance on day and night basis to monitor, inspect and enforce environmental regulations more effectively. Equipped with advanced thermal imaging technology, these drones can detect emissions, illegal discharges and other violations that are hard to observe at night.

  vii. **Mandatory Rainwater Harvesting Systems:** One of the most important water conservation initiatives undertaken was the introduction of regulations mandating rainwater harvesting systems in new construction

projects. Developers of residential, commercial and institutional buildings were required to integrate rainwater collection and recharge mechanisms into project designs through environmental approvals, promoting groundwater recharge, reducing runoff and supporting climate-resilient urban development.

viii. **Water Recycling and Reuse System in Commercial Operations:** Reinforcing a holistic environmental management approach, EPA mandated installation of water recycling and reuse systems in water-intensive facilities such as car wash stations. The initiative result reducing freshwater consumption in the commercial sector and encouraged businesses to adopt sustainable practices.

ix. **Fuel Quality Testing Laboratories:** The EPA has for the first time in Punjab introduced fuel testing laboratories to monitor and ensure the quality of petrol and diesel sold at petrol pumps across the province. These laboratories test fuel samples for adulteration and control environmental pollution and protect vehicle engines. In addition, EPA Punjab has launched mobile fuel testing labs, enabling on site testing and swift enforcement actions against non-compliance fuel outlets, marking a significant step toward improving fuel and air quality across the province.

Sindh

- Regarding, the quality of drinking water, the province-wide assessment conducted by the Sindh Environmental Protection Agency (SEPA) reflects ongoing provincial efforts in ensuring safe drinking water standards across Sindh. A total of 950 drinking water samples were collected and analyzed across all six divisions of Sindh. Of these, 734 samples (77.3 percent) were found fit for drinking purposes, while 216 samples (22.7 percent) were declared unfit.
- In compliance with environmental pollution control measures, SEPA through Environment, Climate Change & Coastal Development Department has implemented province-wide interventions. An updated survey of wet process industries shows that Karachi Division has 509 wet process industries, of which more than 174 have installed wastewater treatment plants, with approximately 104 million gallon per day (MGD) of effluent generated in Karachi, out of which 30 MGD get treated.
- Regarding incineration plants, SEPA has approved 18 hazardous waste incineration plants to reduce waste volume, destroy hazardous components and mitigate methane emissions from landfills.
- For vehicular emission monitoring, continuous monitoring has been conducted in coordination with traffic police. A total of 37,977 heavy vehicles were monitored, out of which 10,858 vehicles were challaned or penalized. Moreover, SEPA has directed all industries to submit quarterly vehicular emission monitoring reports.
- Under targeted actions to control air pollution:

  i. Operations of 218 illegal battery recycling units have been stopped and demolished across Sindh province, mainly operated in Karachi and Hyderabad.

  ii. SEPA has also enforced the installation of Zig Zag technology in 10 brick kiln units and 83 units were prosecuted. The majority of brick kilns using hazardous fuels such as used tyres, shoes and plastic have been stopped from their operations.

  iii. Further, all illegal coal storage sites were removed, and coal was shifted to

legal storage yards particularly located in district Malir.

iv. Actions have also been taken against all stone crushing units in Sindh province for controlling dust emissions; they have been directed to implement SEPA guidelines and environmental management plans. As an update, in Larkana region, 147 rice mills have installed air emission control systems (cyclone separators/dust catchers). In Mirpurkhas region, 103 cotton ginners have installed dust catcher systems. Monitoring of cement industries in Sindh has been completed and found that all cement factories are in compliance with air pollution control equipment requirements.

- Regarding environmental development projects:

  i. The Jam Chakro Landfill Site is being upgraded from a traditional open dumping area into a modern engineered sanitary landfill. The project envisages potential utilization of recovered landfill gas for power generation, with an estimated annual GHG emission reduction of approximately 0.4 million metric tons of CO_2 equivalent ($MtCO_2e$).

  ii. The Green Bus Rapid Transit system in Karachi, a 21.7-kilometer route from Municipal Park to Surjani Town, is projected to reduce carbon emissions by 2.6 million metric tons of CO_2 equivalent over its 30-year life. SEPA has approved the Environmental Impact Assessment of the Green Line BRT and the Red Line BRT (under construction), while the Yellow Line BRT is under review.

  iii. With regards to renewable energy, SEPA has issued approvals to 35 wind energy companies producing total of 1,845 MW. A single megawatt of wind power can avoid approximately 2,400 tons of CO_2 emissions per year. Similarly, approvals have been issued to 5 solar power projects producing 350 MW, with one megawatt of solar power avoiding approximately 1,400 tons of CO_2 emissions per year.

- Regarding pyrolysis plants, the Government of Sindh imposed a ban on tyre pyrolysis plants due to unscientific practices and lack of environmental controls. Subsequently, more than 20 such plants were shut down.

- Lastly, with regards to environmental legislation, the Sindh Cabinet approved the Environment, Climate Change & Coastal Development Department's proposal regarding ban on substandard fuel for the prevention and control of hazardous emissions.

Khyber Pakhtunkhwa

- Khyber Pakhtunkhwa Wildlife Department took significant initiatives to strengthen biodiversity conservation in the province. During July–December 2025, Rs 533.6 million (US $ 1,913,842) revenue has been generated inducing exportable Markhor permits worth Rs 263.7 million (US $ 946,000) from the bid of 39 Big Game Hunting Permits. Further, 1,647 small game and 9 big game licenses have been issued. Similarly, under non-exportable quota under Big Game-Hunting, 20 Ibex and 6 Grey-Goral permits were successfully bid out, valued at US $ 16,042 and US $ 398,500, respectively

- Likewise, corresponding to broader efforts to promote climate resilience and biodiversity conservation, Darsam and Hangu and Lower Tanawal Beer Valley in Abbottabad arid Haripur have been notified as Community Game Reserve and Gallies Abbottabad, Chitral Bashkar Garamchashma have been declared as Biosphere Reserves, under the categories of protected areas.

- Regarding forestry initiatives in the province, under the revised 10 Billion Tree Tsunami Programme (10-BTTP) significant progress has been made with the plantation of 0.3 million plants across all categories and the free distribution of 0.2 million farm forestry plants during July–December 2025.
- Additionally, the update as of July–December 2025 on Forestry Sub-Sector project include establishment of 20 Seed Stands in Merged Areas (10 in Chilghoza Forest, 2 in Oak Forests, 4 in Acacia Forests and 4 in other Forest "Gurgura, dodonea, Chir & Kail"). Additionally, 4 Bio-Gas Plant has been established in South Waziristan in order to provide alternate source of energy and to reduce pressure on Chilghoza Forests.
- Lastly, the updated status (July–December 2025) of NTFP Sub-Sector project comprise of: Provision of bio-briquette technology to 135 local communities, Value chain development of one Chilghoza product initiated, Propagation of lemongrass was carried out over 2 acres, 165 apiculture kits and 228 medicinal plant collection kits distributed, Walnut nurseries in Malakand and Hazara established over 4.2 acres, In-situ conservation areas covering 47 acres created, 188 Chilghoza cone collection tools distributed, 50 acres of degraded Chilghoza forests conserved and restored and Non-timber Forest Product nursery has been established on 0.5 kanal land.

Balochistan

- The implementation update for the use of Plastic Shopping and Flat Bags Act, 2023 reflects its continued progress across Balochistan. The Act has been promulgated and enforced province-wide, alongside mass awareness seminars and stakeholder consultations for its effective implementation. For sound management of plastic (polythene) pollution, the Balochistan Environmental Protection Agency (BEPA) has confiscated approximately 158,293kg of illegal plastic shopping bags from July–March 2026.
- The establishment of a dedicated Climate Change Unit within the Planning & Development Department, Government of Balochistan, marks further progress in strengthening institutional arrangements for climate action. The unit has been set up on a priority basis to address the growing need for integrated climate action and sustainable development, focusing on policy formulation, project planning and resource mobilization for climate mitigation and adaptation initiatives.
- The update on the installation of Effluent Treatment Plants (ETPs) indicates continued progress in industrial pollution control in the province. Effluent generated from industries such as textiles, pharmaceuticals, chemicals and tanneries is being managed through the installation of ETPs to prevent contamination of water resource. In the Hub industrial area, out of 34 industrial units generating effluent waste, 26 have installed ETPs while 4 other are currently in the process and the remaining 4 plans for the installation.
- The update on hospital waste management reflects strengthened regulatory enforcement under the updated Balochistan Hospital Waste Management Rules, 2025. Regular monitoring and inspection of 14 public and 49 private hospitals in Quetta are being conducted, alongside strict surveillance and enforcement actions (such as legal proceedings against violators) by the BEPA. As a result, a total of 195,980.3 kg of infectious hospital waste was incinerated during the period July–March 2026.
- The update on actions against brick kiln emissions reflects continued provincial efforts to transition from Bull Trench Kilns (BTKs) to Zig-Zag Kiln (ZZK) technology. Approximately 353 brick kilns are currently operational, of which 40 have been converted to ZZK technology, while more

than 250 kilns have been relocated to remote areas of District Mustang to mitigate environmental impacts.

- Further, to strengthen public-private collaboration for environmental sustainability in the province, a Memorandum of Understanding (MoU) has been signed between the Lasbella Chamber of Commerce and BEPA. Under the agreement, industrial units will collectively undertake the plantation of 1,000 trees annually as part of efforts to mitigate environmental hazards and enhance green cover in the province.
- The update on safeguarding the coastal environment of Balochistan highlights ongoing efforts to control marine pollution through improved industrial effluent management in the Hub Industrial & Trading Estate (HITE) and surrounding industrial clusters. At present, 160 industrial units in HITE and 25 marble units in District Hub are operational, of which 32 units discharge industrial effluents. In this regard, 4 units have already installed Effluent Treatment Plants (ETPs), while installation in the remaining 2 units is in progress. As a result, a total of 900,000 gallons of wastewater are being treated and safely discharged into the system in accordance with prescribe standards.
- Additionally, the province has made substantial progress towards meeting the compliance cut-off date of June 2025 under Pakistan's 2023 accession to the Hong Kong International Convention (HKC) on ship recycling. In this regard, safe and environmentally sound recycling of ships at Gaddani has been focused on. Gaddani comprises 132 ship-breaking yards operated by 25 shipbreakers, contributing significantly to steel production, tax revenue and employment (directly and indirectly supporting a dependent population of around 3.5 million). As an update to this initiative, 11 shipbreakers have completed the required yard work for HKC compliance and have formally submitted their application for certification.
- Regarding monitoring and regulation of vehicular emissions, the BEPA, in collaboration with relevant departments, identified 16 public transport routes in Quetta and conducted targeted inspections of buses operating on these routes. Following the exercise, route permits were cancelled for vehicles emitting excessive smoke, while compliance stickers were issued to compliant vehicles.
- Furthermore, BEPA has strengthened environmental compliance through continued effective implementation of Environmental Impact Assessments (EIA) and Initial Environmental Examinations (IEE). In this regard, 24 EIAs and 301 IEEs were conducted during July–March 2026, generated significant revenue of Rs 17.45 million.

Concluding Remarks

Pakistan is on the frontlines of climate crisis. Climate change is no longer a future threat but a current reality affecting people, economies and ecosystems, with effects already seen in various sectors. The challenge extends beyond rising temperatures to how effectively and urgently responses are mobilized, as the crisis continues to intensify with far-reaching human and economic consequences. In a rapidly evolving global climate landscape, Pakistan's role is also expanding through active climate diplomacy at international forums such as COP30 to more active participation in climate finance and international partnerships, including being at the forefront at Climate Vulnerable Forum (CVF-V20) and employing avenues such as Resilience and Sustainability Facility, Green Climate Fund, Adaptation Fund and Global Environment Facility. In addition, the country is also pro-actively advancing localized climate agenda through Monsoon 2026 Strategic Plan, Glaciers Preservation and Resilience Initiative, Pakistan

Green Taxonomy, NDC 3.0 and Pakistan Climate Prosperity Plan.

The Government of Pakistan has remained committed to addressing climate change, with environmental sustainability being at the forefront of national climate priorities. Climate change has increasingly become an integral part of Pakistan's mainstream economic and governance agenda, supported by sustained policy direction towards adaptation and mitigation, increasing inter-provincial and federal coordination. Pakistan is transitioning from recognizing climate change as a systemic risk toward strengthening climate resilience, however, the scale and urgency of the challenge demand multifaceted and strong implementation, greater and effective financial mobilization and sustained policy continuity. Pakistan is now at a pivotal stage in its climate change journey. The momentum generated through national and international efforts needs to be translated into effective implementation through sustainable holistic and forward-looking approach, policy effectiveness, institutional strengthening and enhanced global cooperation, thereby securing a resilient and sustainable future of the country.